

On the database and its effects on how we think about, and occupy, a space artistically

Sobre Database e seus impactos na forma de pensar, e ocupar, artisticamente um espaço

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Abstract

The concern of this article is to set out what the database and algorithms would mean as media that alter the ways of thinking about, and acting upon, space — both in the form of an awareness and an analysis of trends, Lev Manovich's "digital humanities", and at the level of any user who interacts and creates artistic expressions while walking through the streets in the course of everyday life. The texts and projects by Manovich presented here, such as the data-visualisation project called "Phototrails", and the artists discussed, who use social media to circulate works made in public spaces, all draw on the same service, Instagram, which has already become the most widely used in the world for quick posts consisting solely of photographs and videos of up to 15 seconds. When one is dealing with the Computer Age at the level of its impact upon world view, upon symbolic forms, it is never enough to speak only of what is visible exclusively, or principally, through a screen or other hardware.

Keywords: database, algorithms, Instagram, public art

Abstract (the author's own English, as printed in the original)

The main issue of this article is to present what does it mean database and algorithms as mediums that change the ways of thinking about space. Considering a consciousness and trend analyses, but also in the level of any user who interacts and create artistic expressions when he walks daily on the streets. The texts and projects present from Lev Manovich, such as the data visualization project called "Phototrails", and the other artists who use the social media to show their works made on public space, all of them use the same service, Instagram, which is already the most used when it comes to fast photo and 15 seconds videos sharing. When one talks about the Computer Age at the level of its impact in terms of world view, of symbolic forms, its never enough to speak only about what is visible exclusively, or as its main source, through a digital screen and other hardware.

Keywords: database, algoritm, instagram, public art

1. Introduction

In 1998 Lev Manovich, a Russian based in the United States, published his article "Database as symbolic form", in which he argues that data structures and algorithms are two halves of the

ontology of the world according to a computer.¹ This would imply, according to the author, that the computerisation of culture involves the projection of these two fundamental parts of software onto the cultural sphere. Following the text, if the novel was the great cultural production of the nineteenth century and cinema that of the twentieth, in the Computer Age the database would be the principal cultural production; there is, however, the particularity that it does not tell a story but presents collections of individual items, in which each one has as much meaning as any other — and yet neither are they simple collections of items: they possess their own, and varied, capacities for self-organisation and they interact continuously with algorithms.

Manovich compares the database with other cultural productions, saying that each presents a different model of what the world is, and in this sense he takes as his reference the concept of symbolic form, citing the famous essay by Erwin Panofsky. That concept makes its first appearance in the work of Ernst Cassirer, a German theorist of the late nineteenth and early twentieth centuries who is of the greatest importance for having connected the philosophy of culture to that of the history of science, and who was also an essential figure in the consolidation of the latter as a scientific discipline. Cassirer's first works of intellectual weight were bound up with the philosophy of mathematics and of natural science, but in the 1920s, on coming into contact with the Library of Cultural Sciences founded by Aby Warburg in Hamburg, he began to turn his understanding towards what was then called the science of culture. The German theorist sought to bring into his studies of culture part of the methodology and the understanding proper to the natural sciences, but also to philosophy — both analytic philosophy and thinkers such as Kant and Hegel — weaving a theory of how the "facts of science" may form part of the "facts of culture". He discusses how the transformations of mathematical theories of space and time cannot be adequately understood without attending to their more concrete and intuitive symbolic manifestations: for example, how a space is represented within a cultural context.²

The same concept is presented by Panofsky in his essay "Perspective as Symbolic Form", where it is described as a "spiritual meaning attached to a concrete, material sign and intrinsically given to this sign",³ drawing directly upon Cassirer's research, in the very decade of its publication. That work of Panofsky's is a major point of reference in twentieth-century cultural theory, constantly cited, because it succeeds in revealing the extent to which modern perspective is the particular creation of a civilisation, grounded in a very particular world view, a way of understanding physical space systematically. The most important thing about his essay is precisely that it reveals the extent to which our understanding of space is not the fruit of a natural and continuous perception, but is culturally specific. The conception established by perspective refers to a continuous, homogeneous and infinite space, something essential to modernity, which would be clearly expressed in the philosophy of Descartes: the creation of a rationalist and individualist gaze, one that places the world under the sight (and the control) of individual perception, inaugurating an anthropocratic period, one more autonomous in relation to religion. Our relation to reality is therefore profoundly altered according to the symbolic forms that are familiar to us.

2. Development

When Manovich treats the database as a symbolic form, this means that the principal axis of his theoretical argument is the impact of the computer's logic upon the world view of contemporary society. Manovich is a prolific author, with several new articles each year — at least ninety between 1993 and 2016 available on his website — besides having produced a number of books. Since the publication of "Database as symbolic form" he has become ever more interested in what he calls "digital humanities", arguing that "data science" has become the eye necessary to make cultural productions intelligible at the scale of the information storage available to us today.⁴ There are various articles on the nature of software, on the social trends perceived from the posts of millions of people on Instagram, on the principles and possibilities of data-visualisation processes, and so on. As early as 1998, in the article first cited here, the author was already pointing out that today's media would require a new aesthetics.

Following art historian Ervin Panofsky's analysis of linear perspective as a "symbolic form" of the modern age, we may even call database a new symbolic form of a computer age (or, as philosopher Jean-Francois Lyotard called it in his famous 1979 book *Postmodern Condition*, "computerized society"), a new way to structure our experience of ourselves and of the world. Indeed, if after the death of God (Nietzsche), the end of grand Narratives of Enlightenment (Lyotard) and the arrival of the Web (Tim Berners-Lee) the world appears to us as an endless and unstructured collection of images, texts, and other data records, it is only appropriate that we will be moved to model it as a database. But it is also appropriate that we would want to develop poetics, aesthetics, and ethics of this database.⁵

Manovich's research, above all from the 2000s onwards, then came to focus its theoretical study and its proposals for an aesthetics of artistic expression upon data visualisation. In 2007 he founded the Software Studies Initiative, based in New York and in La Jolla, California, with which many of his works and research projects grounded in data visualisation are connected.

Fig. 1 — Visualisation of Instagram images, organised by colour. Part of the Phototrails project. Source: www.manovich.net

Phototrails is a research project by Manovich that uses media-visualisation techniques to explore visual patterns, dynamics and structures in photographs shared by users. Using the APIs of the most popular image-sharing services, such as Instagram, the researchers downloaded millions of public images and their metadata. They analyse each image and then visualise groups of images by means of their own software tools, organising them according to criteria of their own, such as colour, time of day, and characteristics of the users (for example, whether they live in the city or are visitors). In this project there were 2.3 million Instagram photographs shared by people in thirteen global cities. This is one example of data visualisation among many other projects with similar aims by the Russian-American. In each project, different possibilities for organising the data and different forms of visualisation and interactivity are explored. There are also applications released by the Software Studies Initiative so that users may participate more actively, or learn about the analyses that can be made from the recognition of visual patterns.

Considering, however, the question of the "database" as a symbolic form in its full breadth, would it not be pertinent to study how far it affects our thinking about space and creates aesthetics of its own beyond the digital medium? If we think of the multifaceted observer who is reader, spectator and internet user, someone who simply walks in the street and occupies the

public space of the city, the aesthetics, ethics and poetics that result directly from the impact of the computer age extend far beyond screens.

According to Manovich himself, in a more recent article of his, from 2016, entitled "Social Media Inequality: Definition, Measurements, and Application", the social-media content shared today, such as Instagram images with their tags and descriptions, is the key form of life in the contemporary city: "It tells people where activities and locations that interest them are and it allows them to share their urban experiences and self-representations. Therefore, any analysis of urban structures and cultures needs to consider social media activity."⁶

It would be mistaken to separate what is happening in the streets from its virtual counterpart. As Néstor García Canclini stresses with regard to readers, spectators and internet users, these cultural receivers are concentrated in one and the same person, and the cultural industries are uniting languages and combining spaces in a technological process of digital convergence. While it might be argued that this was already happening at an ever more accelerated pace through devices such as radio and television, with the spread of the smartphone and of digital platforms — with countless new applications released every day, each connected to the others by way of a single access account — the process seems to take a different direction, that of the database and the algorithms discussed by Manovich. It is striking to observe how the organisation of political demonstrations, of occupations of public space, has come to be coordinated through virtual platforms in recent years, even in countries outside the western centre, as was the case with the Arab Spring in 2011.

The first idea is that public attention is steered by the attentions of the media—which means images. When there are photographs, a war becomes "real." Thus, the protest against the Vietnam War was mobilized by images. The feeling that something had to be done about the war in Bosnia was built from the attentions of journalists: "the CNN effect," it was sometimes called, which brought images of Sarajevo under siege into hundreds of millions of living rooms night after night for more than three years. These examples illustrate the determining influence of photographs in shaping what catastrophes and crises we pay attention to, what we care about, and ultimately what evaluations are placed on these conflicts.⁷

Susan's text seeks to make clear how far the manner in which images are shared influences the reaction of millions of spectators. She argues in several of her books, including the acclaimed *On Photography*, that the mass reaction of protest against the Vietnam War was in large part the result of its having been the first televised war, with rapid — or even live — sharing of images of the fighting. The author maintains, however, that the mere sight of images of horror is not enough to provoke a political consciousness: this must come first; there must already be an empathy with the victims in order to feel their pain.⁸ Although we are bombarded with images in the contemporary city, they still do not speak for themselves.

If this was already visible in the 1970s, what is to be said of today's possibilities for sharing images? As the Arab Spring, the Occupy movement and the Brazilian demonstrations of 2013 and 2016 have shown, virtual platforms are today part of public space — both in the sense of coordinating the occupation of, and the life lived within, physical space, and in the sense of a space of public debate. One cannot even make an accusation against a person on a personal Facebook profile without being aware that this may give rise to legal proceedings. And just as public space is not simply democratic, in the sense of being equally open to occupation by all,

having its countless codes and censorships of its own (one need only think of the action of the police and of the countless pieces of evidence of racist behaviour on their part in a number of different countries), neither is digital space free of filters and censorship (such as the Facebook and Google algorithms that present content to a user on the basis of what has been perceived to be to their taste, or even directly political matters, such as Facebook's preventing Syrian refugees from adding American citizens as friends).⁹

Returning to the question of the expressive, aesthetic, ethical and poetic possibilities arising from the Computer Age that Manovich raises: how far might the new media affect artistic occupation and expression in public space? The editors of the website widely recognised for publicising works with an artistic intent in public space, the Wooster Collective, founded in 2001 by Marc and Sarah Schiller, write in the book *Trespass: A History of Uncommissioned Urban Art* that technology played a crucial role in the rapid development of unauthorised practices of public art in the 2000s. The affordably priced digital camera, for example, made it possible to record and to share any work, just as the internet allowed artists to see works from all over the world in a single night in front of the computer. In the exchange of stories, criticism and techniques, a strong community developed, a community that drives these artists' experimentation and refinement. If these are matters common to any working group today, for Marc and Sarah, in the case of the artists who intervene in public space belonging to the most recent generation — one that came of age during the rise of the internet — it is a strong sense of community, the fruit of that contact with others over the web, that characterises the group.

Or, as these authors argue in interviews: "Street art is truly the first global art movement fuelled by the internet. (...) The rise in popularity and acceptance of street art are directly connected with the rise of the internet."¹⁰ Even if the designation "global art movement" is questionable, the Wooster claim seems pertinent, and it points to a process of legitimation and popularisation of the more recent practices of art in the street that is quite different from those experienced, for example, by what was called the graffiti movement in the 1970s, or by Mexican muralism in the 1930s — both of which were widely publicised by the print media.

For a good part of the art produced in public space, above all the paintings, these websites were its shop window and its connector long before sponsored urban-art festivals began to proliferate (such as Nuart in Norway, Fame in Italy, Wynwood in Miami and countless others around the world, Brazil included), and long before the doors of the great art institutions were opened to it. Few artists coming out of the illegal urban-intervention scene had achieved recognition in terms of sales and prices before the 2000s; after all, the graffiti boom of the 1980s was itself short-lived, as the artist Futura2000 recounts.¹¹ These artists today produce in the knowledge of that large international community of spectators which will judge their works through photographs and videos.

Fig. — Nobody Likes Me. Intervention published on Instagram by the artist himself. Found on the site: <http://thesource.com>

Lev Manovich also points out, in another article, "How to Follow Software Users", published in 2012, that between 2004 and 2006 there was a turn on the internet, since the most prominent pages ceased to be relevant for their content and became relevant instead for being platforms for exchange between users, opening the way to what we today call social media.¹² Once again,

for the worldwide phenomenon of the acceptance and popularisation of works made in public space discussed by Marc and Sarah, even the illegal ones, Instagram and Facebook are the principal galleries for the sharing of images.

3. Conclusion

This text does not argue that the presence of a good work is simply replaceable by its image on a computer screen or in print; but could it be said that all these works have their primary fruition in their encounter in the street? The countless possibilities for the digital fruition, by way of images, of works made specifically for a given space are perhaps not so very different from Walter Benjamin's interest in cinema as an art based upon technical reproducibility, in which the idea of an original piece or film is no longer so relevant.¹³ The contact of a contemporary observer carrying a smartphone, for example, with a work of art, or with images in general, is always already mediated by the possibility of turning anything at any moment into a photograph or a video, and even of sharing these simultaneously on a virtual social network.

As Jonathan Crary says in his book *Techniques of the Observer* with regard to the camera obscura in the eighteenth century and the stereoscope in the nineteenth, the observer and the various technologies bound up with vision should not be understood as wholly distinct entities with separate identities. Our visual understanding is always shaped by the apparatuses that create and transmit images, as well as by the epistemological conceptions of an epoch — ideas Crary exemplifies when he stresses the impossibility of a theory of pure visibility in the eighteenth century. We would then be living, in the twenty-first century, through another period of transformation in the modes of looking, in so far as the means of producing and of seeing images are undergoing rapid change — and here it is worth recalling Flusser's concept of zero-dimensionality, which underscores the epistemological rupture introduced by the pixel. Through these urban interventions circulated on social media, or through the analysis of the images of social media themselves by means of data visualisation, one may think the possibilities of imagining and of acting in the city today, showing trajectories that point to a relation between work and space very different from that of, for example, Mexican muralism — such as the large, planned paintings made in open parks by Diego Rivera — or that of public monuments, produced under government commission and generally tied to the institutionalisation of the city's memory; both examples rest upon a fruition that is above all the observer's bodily presence. Technologies such as those that create databases are responsible for an alteration in the way of thinking, producing and enjoying images that irrevocably alters our everyday life in its most ordinary actions, such as walking about with no particular purpose.

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Notes

¹: This could be interpreted both by taking ontology as a philosophical tradition and in terms of its practical application in computer science, as a data model representing a set of concepts within a domain and the relations between them.

²: According to the Stanford University Dictionary of Philosophy, found at the link <http://plato.stanford.edu/entries/cassirer/> — accessed June 2016.

³: CASSIRER, Ernst, apud PANOFSKY, Erwin. — the author's own translation — "colocar página etc." [an unfinished note left standing in the original: "insert page etc."].

⁴: MANOVICH, Lev. "Data Science and Digital Art History".

⁵: MANOVICH, Lev. "Database as symbolic form". [The passage is given here in Manovich's original English, which the author reproduced in this note; the body of the original article carried her own Portuguese translation of it.]

⁶: MANOVICH, Lev. "Social Media Inequality: Definition, Measurements, and Application". [The passage is given here in Manovich's original English, which the author reproduced in this note; the body of the original article carried her own Portuguese translation of it.]

⁷: SONTAG, Susan. "Looking at War: Photography's view of devastation and death", published in 2002 in the New Yorker. <http://www.newyorker.com/magazine/2002/12/09/looking-at-war>

war — accessed June 2016. [The passage is given here in Sontag's original English, which the author reproduced in this note; the body of the original article carried her own Portuguese translation of it.]

⁸: SONTAG, Susan. "Regarding the pain of others",

⁹: The author of this text witnessed the moment when a Syrian refugee in Rio de Janeiro demonstrated that he could not send a friend request to American users through Facebook, in December 2015. Facebook prevented him from sending it, but the American woman present at the time was able to do so. The same did not happen when the Syrian added Brazilians to his profile.

¹⁰: SCHILER, Marc and Sara. In "Beyond the streets". [The passage is given here in the Schillers' original English, which the author reproduced in this note; the body of the original article carried her own Portuguese translation of it.]

¹¹: Interview published in the book "Beyond the Street"; see bibliography.

¹²: MANOVICH, Lev. "How to Follow Software Users".

¹³: BENJAMIN, Walter. "A obra de arte na era de suas técnicas de reprodução" [The Work of Art in the Age of Its Technological Reproducibility]. São Paulo: Abril, 1983.